

marking its transition from a research entity to a functioning semiconductor enterprise. These early prototypes targeted high-frequency GaN chips, particularly in telecom base stations and defence radar systems, where high power density and efficiency are critical.

“Our first focus was to prove that India could build world-class GaN RF devices,” says Chandrasekar. “Once that was validated, the focus shifted to scaling.”

AGNIT’s core product line today includes:

- **GaN epitaxial wafers.** Custom-grown GaN films on silicon and silicon carbide

- **GaN RF devices.** High-efficiency transistors and amplifier modules for telecom, radar, and aerospace applications, covering frequencies from S-band to X-band

- **Evaluation boards and modules.** Ready-to-use boards that help customers test and integrate AGNIT’s devices faster

In December 2023, AGNIT signed a contract under the iDEX (Innovations for Defence Excellence) programme of India’s Ministry of Defence. The partnership supports the development of indigenous GaN components for radar and electronic warfare systems.

“For decades, silicon has been the default semiconductor material. However, GaN offers distinct physical properties. It provides about five times higher electron



Hareesh Chandrasekar, Co-founder, AGNIT Semiconductor

mobility and can withstand roughly three times the voltage of silicon before failing,” he informs.

It conducts electricity very efficiently when switched on and blocks high voltage when switched off. At the same time, it conducts heat more effectively and can be built in a small, compact form.

AGNIT runs its production inside the IISc facility, using a small-volume GaN foundry supported by the Ministry of Electronics and IT (MeitY). It allows startups to access advanced fabrication without the massive investment required for commercial labs.

“When people tell you they need 20 billion dollars to set up a silicon fab, a team could spend 500 million dollars and set up a reasonably sized gallium nitride fab,” Hareesh Chandrasekar discloses.

Manufacturing GaN chips differs significantly from manufacturing silicon chips. Native GaN wafers are extremely difficult to make compared to silicon substrates, so GaN is typically deposited on silicon wafers, sapphire wafers, or silicon carbide wafers. LEDs often use sapphire because it is transparent.

“Our RF components are built on silicon carbide due to its thermal conductivity. Fast chargers rely on GaN on silicon because it is cost-effective,” he says.

Funding and talent remain key hurdles. Developing semiconductors requires patient capital, often for years before revenue is generated. “There is not much VC money for semiconductors. Grants and government funds become essential,” Chandrasekar says. The company also faces the challenge of building specialised expertise in India, where most semiconductor jobs traditionally centre on design services for multinationals.

Parallel to the RF chips, AGNIT is pushing its GaN product segment beyond defence and telecom into power electronics. The team is experimenting with GaN-powered EV chargers for their compact and fast-charging capabilities. “We are still testing; the potential is massive. If this scales, the next EV charger could fit in a backpack, smaller, faster, and far more efficient than anything silicon can do today.”

—Janarthana Krishna Venkatesan

3 “WE TURN FACTORIES INTO SMART DATA HUBS THAT SPEAK BACK”

This startup has built IoT devices and AI solutions that enable small factories to monitor every machine and process, turning complex operations into clear, real-time insights.

The Covid era changed many lives. Three brothers, Ananth Kulkarni, Vaman Kulkarni, and Kiran Kulkarni, from diverse fields, came together to address the challenges faced by medium-sized

enterprises (MSMEs). They named the startup Hexitronics, inspired by the six-sided geometry of a honeycomb.

The startup designs and manufactures IoT devices that now serve as key data sources for AI engines.

Its solutions cover multiple use cases, primarily for small- and medium-sized enterprises such as rice mills and stone crusher units. These businesses often need remote monitoring of operations, and the devices enable them